THE PROJECTION FORMULA MAP IS AN ISOMORPHISM FOR A MORPHISM OF RINGED SPACES THAT IS A HOMEOMORPHISM ONTO A CLOSED SUBSET

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites six cross-chapter labels, including the equation defining the projection formula map; each is recorded in assembly.external_references with its canonical tag as a navigation pointer.

Editorial note. The proof cites no bibliography item inside the counted span.

Allowed standard prerequisites (editorial).

- Exactness of pushforward along a homeomorphism onto a closed subset. Stacks Project, Modules on Sites, Lemma modules-lemma-i-star-exact (tag 01AX). Allows replacing Rf_* by f_* on any representative complex, which is what makes the comparison a levelwise computation.
- K-flatness and the projection formula for pullback. Stacks Project, Cohomology of Sheaves, Lemma lemma-pullback-K-flat (tag 06YC). Supplies a representative of Lf^*K with which the tensor product computation can be performed on the level of complexes.
- Stalks of closed pushforward and of pullback of modules. Stacks Project, Sheaves, Lemmas sheaves-lemma-stalks-closed-pushforward (tag 00AE) and sheaves-lemma-stalk-pullback-modules (tag 0098). Identify the stalks appearing in the final computation of the single-module identity.

Source reference. Original equation 0B53: sources/stacks-chow/tags.txt, source lines 14021–14021. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01AX: sources/stacks-chow/tags.txt, source lines 1706–1706. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 06YC: sources/stacks-chow/tags.txt, source lines 8881–8881. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01AZ: sources/stacks-chow/tags.txt, source lines 1708–1708. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00AE: sources/stacks-chow/tags.txt, source lines 386–386. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Date: stacks-project:cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original reference 0098: sources/stacks-chow/tags.txt, source lines 344–344. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of ringed spaces such that f is a homeomorphism onto a closed subset. Then (0B53) is an isomorphism always.*

Proof. Since f is a homeomorphism onto a closed subset, the functor f_* is exact (Modules, Lemma 01AX). Hence Rf_* is computed by applying f_* to any representative complex. Choose a K-flat complex $\mathcal{K}^\bullet$ of $\mathcal{O}_Y$-modules representing K and choose any complex $\mathcal{E}^\bullet$ of $\mathcal{O}_X$-modules representing E. Then Lf^*K is represented by $f^*\mathcal{K}^\bullet$ which is a K-flat complex of $\mathcal{O}_X$-modules (Lemma 06YC). Thus the right hand side of (0B53) is represented by

$$f_*\mathrm{Tot}(\mathcal{E}^\bullet \otimes_{\mathcal{O}_X} f^*\mathcal{K}^\bullet)$$

By the same reasoning we see that the left hand side is represented by

$$\mathrm{Tot}(f_*\mathcal{E}^\bullet \otimes_{\mathcal{O}_Y} \mathcal{K}^\bullet)$$

Since f_* commutes with direct sums (Modules, Lemma 01AZ) it suffices to show that

$$f_*(\mathcal{E} \otimes_{\mathcal{O}_X} f^*\mathcal{K}) = f_*\mathcal{E} \otimes_{\mathcal{O}_Y} \mathcal{K}$$

for any $\mathcal{O}_X$-module $\mathcal{E}$ and $\mathcal{O}_Y$-module $\mathcal{K}$. We will check this by checking on stalks. Let $y \in Y$. If $y \notin f(X)$, then the stalks of both sides are zero. If $y = f(x)$, then we see that we have to show

$$\mathcal{E}_x \otimes_{\mathcal{O}_{X,x}} (\mathcal{O}_{X,x} \otimes_{\mathcal{O}_{Y,y}} \mathcal{F}_y) = \mathcal{E}_x \otimes_{\mathcal{O}_{Y,y}} \mathcal{F}_y$$

(using Sheaves, Lemma 00AE and Lemma 0098). This equality holds and therefore the lemma has been proved. □